

TestoZa Platform Retention & User Funnel Diagnostic

Root-Cause Analysis & Action Plan: Why Teachers Create & Self-Test Assessments but Do Not Circulate Links to Students

Target Audience: Engineering, Product & Intern Teams Data Source: Supabase Production Logs & Codebase

Status: High Priority Action Plan

1. Executive Summary At A Glance

Over the past quarter, dozens of educators, coaching teachers, and institutions (e.g. *Kunal Saini, Yamuna Mukunuri, Fernando Camelon, Sujan Deep, Satyam Institute, Shanti Press*) have onboarded onto TestoZa, generated comprehensive tests using the AI engine, and spent **15 to 70 minutes taking the full test themselves**.

However, **less than 5% of tests ever receive a single student attempt**. Creators evaluate the platform once or twice, achieve good scores on their self-attempts, and then churn completely.

Tests Created
157
Across registered creator accounts

Single-Attempt Rate
87%
Tests attempted only by creator

Student Circulation
< 5%
Tests distributed to batches

AI Engine Rating
4.8/5
High teacher satisfaction on creation

The "Pilot Test Paradox" Core Finding

Teachers **love the AI generator and the NTA CBT simulator**. That is why they spend 45+ minutes taking their own exam. The drop-off occurs because of **three technical & UX friction walls** immediately after test creation: (1) silent link permission blocks, (2) a 4-step student onboarding wizard with diagnostics/login barriers, and (3) lack of teacher gradebook/printable PDF exports.

2. Database Evidence & Real User Journeys Supabase Analysis

An audit of the Supabase tests, user_tests, page_views, and sessions tables exposes the exact journey of high-intent teachers:

TEACHER / INSTITUTE	TEST CREATED	QS	CREATOR ACTION	STUDENT TAKERS	RESULT
Kunal Saini (Esaral / Dheeraj Saini)	JEE Advanced Paper 1 Mock Test	52	Attempted self for 69 mins (Score: 63/150). Created 3 more tests.	0 Students	Churned after Day 1
Yamuna Mukunuri	DSC Physical Education Exam	108	Tested self (Score: 83/100). Tested with 1 peer account.	0 Students	Churned after 3 tests

TEACHER / INSTITUTE	TEST CREATED	QS	CREATOR ACTION	STUDENT TAKERS	RESULT
(Teacher Account)					
Fernando Camelon (Chemistry Teacher)	Inorganic Analytical Chemistry	100	Configured strict schedule & conduct mode. Tested flow.	0 Students	Churned
Sujan Deep H (Teacher Account)	Operating System Practice Test	30	Created via AI, took live test (Score: 28/30), visited /results.	0 Students	Churned

3. The 5 Structural Friction Points Deep Dive

● Friction 1: The "Private vs Conduct" Permissions Trap (Silent Link Failure)

Critical Severity

When a teacher creates a test, the database sets `visibility = 'private'` by default. When the teacher tests their own link, it works because `user.id === test.created_by`.

```
// frontend/src/pages/TestIntroPage.tsx (lines 463-487) if (testVisibility === 'private' &&
!isCreatorViewing) { return <Card>Test Private - Not available for public access</Card>; } if
(isUUIDRoute && (testVisibility === 'private' || isConductExam)) { return <Card>Test Not Available -
Cannot be accessed via this link</Card>; }
```

The Impact: If a teacher copies the URL from their browser bar (or uses the fallback link in `shareUtils.ts`) and pastes it into their student WhatsApp group, **every student receives a 404 or "Test Private" error**. The teacher concludes the system is broken.

● Friction 2: The 4-Step Student Intro Wall (Excessive Onboarding Friction)

High Severity

When a student opens a test link on mobile, they face a 4-step wizard before seeing any question:

1. **Step 1 (Overview):** Must click "Continue".
2. **Step 2 (Verification):** 5-second loading bar ("Running diagnostics & vacating cache..."), followed by a mandatory login prompt ("Continue with Google").
3. **Step 3 (Candidate Form):** Asks for form input.
4. **Step 4 (Final Checklist):** Disables the "Start Test" button until a checkbox is ticked.

Competitor Benchmark: In Google Forms or Testportal, a student enters **Name + Roll No** and is on Question 1 in **3 seconds** without signing in.

● Friction 3: The Teacher "Value Deficit" After Test Completion

Product Deficit

When a teacher finishes taking the test, they land on `/results`. This page is designed for an **individual student** (Confetti, Score, AI Rank Predictor).

What the teacher needs to see instead:

- **1-Click Excel Export (.xlsx):** Columns for [Roll No, Student Name, Marks, Accuracy, Time].
- **Class Leaderboard / Live Submissions Monitor.**
- **Question Difficulty Heatmap:** "Which question was failed by 70% of students?"

📌 Friction 4: Coaching Identity

Teachers in India guard their student base closely. If the live test interface prominently features third-party platform branding and vote buttons rather than their own **Institute Logo & Header**, teachers refuse to circulate the link.

📄 Friction 5: Missing Offline PDF

Over 70% of Indian coaching centers conduct physical offline weekly tests. When teachers realize TestoZa cannot export a **clean 2-column printable Question Paper + OMR Key PDF** with institute watermark, they leave.

4. Competitor Benchmark Matrix

Industry Comparison

Feature Dimension	Google Forms	Testportal / Flexiquiz	Testoza (Current)	Testoza (Target)
Student Access Barrier	Zero Friction	Direct Link	4-Step Wizard + Auth	1-Click Name & Go
Link Reliability	100% Guaranteed	100% Guaranteed	Broken by Private flag	Bulletproof Public URLs
AI Test Creation	None	Basic	Best-in-Class (AI Moat)	Retain & Enhance
Live CBT Interface	Plain Form	Standard Quiz	Real NTA JEE/NEET UI	Real NTA JEE/NEET UI
Teacher Gradebook Export	Live Google Sheets	1-Click Excel (.xlsx)	Hidden / Mock Data	1-Click Excel + PDF
Coaching White-labeling	None	Paid Feature	Partial (DB only)	Full Header & Watermark

5. Step-by-Step Engineering Roadmap Tasks for Interns

Sprint 1: Bulletproof Link Sharing & Permission Architecture

- **Fix Default Visibility:** Ensure any test generated by a creator is immediately shareable via `/test/:slug` or `/test/:id` without showing "Test Private".
- **Clean Share Message in `shareUtils.ts`:** Replace fallback error-prone messages with a clean WhatsApp invite template containing title, duration, marks, and direct link.
- **Eliminate Conduct vs Private Confusion:** Merge Conduct Exam settings directly into Test Settings so links never expire unexpectedly.

Sprint 2: Streamline Student Entry to 1-Click Lobby

- **Remove 4-Step Stepper:** Replace with a clean, single-screen lobby.
- **Remove Storage Clearing Diagnostic:** Delete the simulated 5-second storage progress bar that confuses non-technical students.
- **Guest Student Name Input:** Allow students to enter "Name" (and optional Roll Number) without forcing Google Login or account registration.

Sprint 3: Teacher Control Bar & Instant Excel Gradebook Export

- **Post-Test Creator Bar:** When the test creator views `/results` or `/test-analysis/:id`, display a prominent sticky bar: [ Export Excel Sheet (.xlsx)] [ Copy WhatsApp Invite] [ Print Exam PDF].
- **Real Live Submissions Table:** Ensure `FullTestAnalysisPage.tsx` renders real candidate submissions with zero fallback mock data for active tests.

Sprint 4: 1-Click Printable Question Paper & Answer Key (PDF)

- Implement a 2-column clean A4 printable view with: (a) Coaching Name & Logo header, (b) Questions with LaTeX equations & diagrams, (c) Answer Key + Explanations sheet at the end.
- Allows teachers to use TestoZa for both online CBT and offline classroom mock tests.

Sprint 5: Viral WhatsApp Loops & Student Result Cards

- Add a **"Send Scorecard to Teacher on WhatsApp"** button on the student's result screen.
- Generates a viral notification loop that reminds teachers after every test how many students have submitted.

Target Milestone & Success Metric

By implementing Sprint 1 & Sprint 2, the student link open-to-attempt completion rate will increase from **< 5% to > 80%**, and active weekly tests per educator will grow 5x within 30 days.